

ROY WANG

Senior Full Stack Engineer | AI-Powered SaaS | React | Node.js | Python | TypeScript | AWS | LLMs | RAG | Microservices | Spring Boot | .NET

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SUMMARY

Senior **AI Full Stack Engineer** with **8+ years** of experience designing, developing, and delivering **AI-powered SaaS platforms, cloud-native applications**, and enterprise software across **Healthcare, EdTech, FinTech, E-commerce**, and **Transportation** industries.

Experienced in building production-ready AI solutions using **Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), AI Agents, Model Context Protocol (MCP), Semantic Search, Intelligent Automation**, and **Workflow Orchestration** to solve complex business challenges.

Strong expertise in full-stack application development using **React, Next.js, TypeScript, Node.js, NestJS, Python, FastAPI**, and **Go**, with hands-on experience building scalable frontend applications, backend services, APIs, and distributed systems.

Experienced in enterprise application development using **Java Spring Boot, C#/.NET Core, Angular, Vue.js**, and **Laravel/PHP** for business applications, internal platforms, API services, and modernization projects.

Proven track record architecting scalable **SaaS platforms, microservices architectures, real-time systems, enterprise APIs**, and **cloud-native solutions** deployed on **AWS** and **Kubernetes**.

Passionate about building intelligent software products by combining modern web technologies, strong engineering practices, cloud architecture, and **Generative AI** capabilities.

TECHNICAL SKILLS

Programming Languages — JavaScript (ES6+), TypeScript, Python, Go, Java, C#, PHP, SQL

Frontend — React, Next.js, React Native, Angular, Vue.js, HTML5, CSS3, Tailwind CSS, Redux Toolkit, Zustand, React Query, RxJS, Angular Material, Material UI, Bootstrap

Backend — Node.js, NestJS, Express.js, FastAPI, Python Backend Services, Java Spring Boot, Spring MVC, Spring Security, Hibernate, JPA, C# .NET Core, ASP.NET Core Web API, Entity Framework Core, Laravel, REST APIs, GraphQL, WebSockets, Microservices, Event-Driven Architecture

AI Engineering — Large Language Models (LLMs), OpenAI API, Claude API, Gemini API, Retrieval-Augmented Generation (RAG), AI Agents, LangChain, LangGraph, Model Context Protocol (MCP), Prompt Engineering, Function Calling, Semantic Search, Embeddings, Vector Search, AI Workflow Automation, Document Intelligence, AI Chatbots, Intelligent Assistants

Databases — PostgreSQL, MySQL, SQL Server, MongoDB, Redis, Oracle, pgvector, Pinecone, Qdrant, ChromaDB, Database Design, Schema Design, Data Modeling, Query Optimization, Indexing, Database Migration, Stored Procedures, ORM Technologies

Cloud & DevOps — AWS (EC2, ECS, EKS, Lambda, S3, RDS), Azure, Docker, Kubernetes, Terraform, GitHub Actions, Jenkins, CI/CD, Linux, Cloud Monitoring, Infrastructure as Code

Architecture & Engineering — System Design, SaaS Architecture, Multi-Tenant Systems, Microservices Architecture, Distributed Systems, API Design, Enterprise Application Development, Cloud-Native Architecture, Event-Driven Architecture, Security Best Practices, Performance Optimization, Scalability Engineering, Agile/Scrum, Git

Security & Authentication — OAuth2, JWT Authentication, Role-Based Access Control (RBAC), Spring Security, ASP.NET Identity, API Security, Application Security, Data Protection

Testing — Jest, Cypress, Playwright, JUnit, Mockito, NUnit, Unit Testing, Integration Testing, API Testing, End-to-End Testing, Test Automation, Code Quality Practices

PROFESSIONAL EXPERIENCE

SENIOR AI FULL STACK ENGINEER | Anagh Technologies Inc | Piscataway, NJ, US

Dec 2023 – July 2026

Designed and delivered enterprise **AI platforms**, intelligent **SaaS products**, and **cloud-native applications** across **Healthcare, EdTech, FinTech, E-commerce**, and **Transportation** domains, building scalable production systems with modern frontend, backend, cloud, and AI technologies.

AI Platform Development

- Architected enterprise AI platforms using **LLMs, RAG pipelines, AI Agents, LangChain, LangGraph, MCP**, embeddings, and **Semantic Search** to automate knowledge management and business workflows.
 - Developed AI copilots, conversational assistants, and multi-agent systems supporting streaming responses, tool/function calling, conversation memory, **RBAC**, and enterprise integrations.
 - Built document intelligence pipelines using **OCR, Vector Databases**, embeddings, and **LLM-based extraction workflows** for intelligent document processing and enterprise search.
 - Implemented prompt management workflows, AI evaluation pipelines, guardrails, and observability solutions to improve AI quality, reliability, and security.
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Frontend Architecture

- Architected scalable frontend applications using **React, Next.js, TypeScript, SSR, SSG, ISR**, and modern component-based architectures.
 - Developed enterprise dashboards, workflow management systems, and AI-powered interfaces using **React Query, Redux Toolkit, Zustand, Tailwind CSS, Material UI**, and reusable component libraries.
 - Built enterprise frontend modules using **Angular, TypeScript, RxJS, Angular Material**, routing, forms, and API-driven architectures.
 - Developed customer-facing and internal applications using **Vue.js, Vue Router**, component-based architecture, and REST API integrations.
 - Optimized frontend performance through lazy loading, code splitting, caching strategies, and automated testing using **Jest** and **Playwright**.
 - Implemented real-time user experiences using **WebSockets** and **Server-Sent Events (SSE)** for AI conversations, notifications, and operational dashboards.
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Backend & Enterprise Application Development

- Architected scalable backend systems using **Node.js, NestJS, Python, FastAPI**, and **Go** supporting AI applications, SaaS platforms, and enterprise workflows.
 - Developed high-performance backend services using **Java Spring Boot** for enterprise modules requiring robust transaction processing, business logic orchestration, REST API development, and system integrations.
 - Built enterprise services using **C#/.NET Core, ASP.NET Core Web API, Entity Framework Core**, dependency injection, and background processing patterns.
 - Designed and implemented secure APIs using **JWT, OAuth2, Spring Security, .NET Identity**, and role-based access control.
 - Developed modular backend services following **Microservices Architecture, REST API Design**, and **Event-Driven Architecture** principles.
 - Built asynchronous processing workflows using **Apache Kafka**, background workers, queues, and distributed processing systems for AI inference, automation, and analytics.
 - Developed and maintained business applications using **Laravel/PHP**, implementing MVC patterns, database migrations, authentication, and API integrations.
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Database & Cloud Architecture

- Designed scalable database solutions using **PostgreSQL, MySQL, SQL Server**, and **MongoDB**, including schema design, indexing, query optimization, and database migration.
 - Developed data access layers using **Hibernate/JPA, Entity Framework Core**, and ORM-based application architectures.
 - Optimized backend performance through database tuning, caching strategies using **Redis**, and API performance improvements.
 - Deployed cloud-native applications using **AWS** services including **EC2, ECS, EKS, Lambda, RDS**, and **S3**.
 - Containerized applications using **Docker** and **Kubernetes**, implementing automated deployment pipelines with **GitHub Actions, Terraform**, and **CI/CD** practices.
 - Implemented centralized monitoring, logging, and distributed tracing using **Datadog** and cloud monitoring solutions.
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Industry Solutions

- Developed AI-powered clinical assistants, healthcare workflow automation systems, and intelligent document processing platforms.
- Built adaptive learning platforms, AI tutoring assistants, and semantic search solutions for education products.
- Delivered financial workflow automation systems, KYC document analysis platforms, and multi-tenant SaaS applications.

- Developed intelligent commerce platforms with recommendation engines, customer analytics, and conversational shopping experiences.
- Built transportation and logistics solutions supporting shipment tracking, route optimization, ETA prediction, and real-time operational monitoring.

Technologies:

React, Next.js, TypeScript, Node.js, NestJS, Python, FastAPI, Go, Java Spring Boot, C# .NET Core, Angular, Vue.js, Laravel, OpenAI API, Claude API, LangChain, LangGraph, MCP, PostgreSQL, SQL Server, MongoDB, Redis, Kafka, AWS, Docker, Kubernetes, Terraform, GitHub Actions

SENIOR FULL STACK ENGINEER | Instrumentl | San Francisco, California

Jun 2021 – Nov 2023

Developed enterprise analytics platforms and **SaaS applications** focusing on scalable web applications, reporting systems, customer engagement platforms, and business automation solutions.

- Built full-stack SaaS applications using **React, Node.js, TypeScript, PHP, Laravel, and MySQL** supporting enterprise users and high-volume workflows.
- Developed scalable backend APIs supporting analytics dashboards, reporting workflows, user management, subscription features, and third-party integrations.
- Enhanced existing PHP applications by introducing modern **Laravel MVC architecture**, reusable services, database migrations, and improved API structures.
- Built interactive analytics dashboards using **React, TypeScript**, reusable components, and modern frontend engineering practices.
- Developed enterprise frontend modules using **Angular** and **Vue.js** patterns for internal business tools and customer-facing applications.
- Implemented AI-assisted content categorization and recommendation features improving search relevance and customer engagement.
- Built cross-platform mobile applications using **React Native** and **Flutter**.
- Introduced automated testing workflows using **Cypress**, improving deployment quality and release confidence.
- Delivered multilingual enterprise platforms serving global customers across multiple regions.

Technologies:

React, Node.js, TypeScript, Angular, Vue.js, PHP, Laravel, MySQL, PostgreSQL, React Native, Flutter, Cypress, REST APIs

FULL STACK ENGINEER | Relecloud | Manhattan, NY

Apr 2018 – May 2020

Developed modern **E-commerce platforms** supporting online retailers with scalable commerce solutions, customer experiences, and business automation systems.

- Built high-performance web applications using **React, Node.js, JavaScript, and PostgreSQL** supporting large product catalogs and multi-currency commerce.
- Developed backend services for product management, inventory systems, order processing, customer management, and payment workflows.
- Built REST APIs and integrated third-party payment gateways, shipping providers, and external business systems.
- Developed frontend modules using **Vue.js** and **Angular** for administrative dashboards, operational tools, and customer-facing workflows.
- Developed backend applications using **Laravel/PHP**, implementing MVC architecture, authentication, database migrations, and business logic.
- Implemented database optimization strategies including indexing, query optimization, and schema improvements.
- Developed AI-assisted product recommendation and intelligent search capabilities improving product discovery and customer engagement.
- Built analytics dashboards providing insights into customer behavior, sales performance, and operational metrics.
- Improved application performance through frontend optimization, API improvements, caching strategies, and database tuning.

Technologies:

React, Node.js, JavaScript, Angular, Vue.js, Laravel, PHP, PostgreSQL, MySQL, Payment APIs, E-commerce Platforms, Performance Optimization

EDUCATION

B.S IN Computer Science | Merrimack College | North Andover

2013 – 2017